

Modeling of Current-Driven End-Effect Force Ripple in Air-Cored Linear Synchronous Motor With Multiple Modular Stators

Jae Hyun Kim , Sang Won Jung , Yoon Sik Kwon , Sangmin Lee , and Jun Young Yoon 

Abstract—This article presents a segmented layer model (SLM) for the air-cored linear permanent-magnet synchronous motors (LPMSMs) with multiple modular stators. In the LPMSM with modular stators, the significant current-driven ripples (CDRs) are observed as the permanent magnet mover moves across the stator modules, due to the magnetic end effects. Such CDRs become more severe as the current excitation level increases. In order to correctly model the overall motor forces in the LPMSM, the analytic motor model needs to be able to capture the magnetic end effects from both the stator and mover, and also include the current-driven effects. The SLM method presented in this article divides the analytical region into several segmented layers to be able to accurately capture both the geometric-saliency- and current-driven force ripples by using layers with different fundamental spatial frequencies. The fidelity of the proposed SLM is validated both by the finite element method (FEM) and the experiments. The SLM achieves an accuracy of more than 97% and 87% compared to the FEM and experimental results, respectively, while reducing the computation time by two orders of magnitude as compared to the FEM.

Index Terms—Current-driven force ripple, end effect, linear permanent magnet synchronous motor (LPMSM), modular stator, segmented layer model (SLM).

I. INTRODUCTION

LINEAR permanent-magnet synchronous motors (LPMSMs) have been widely used in various industries such as high-precision position control systems and photolithography machines due to their high force density and

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conversion efficiency [1], [2]. While rotary motors and linear motors are operated by the same electromagnetic principle, the linear motors have magnetic end effect due to their finite length [3]. The nonperiodicity of magnetic field distribution at both ends of the linear motors contributes to the force ripple, leading to undesired vibration and reduced accuracy of position control in servo systems [4], [5]. Such magnetic field distortion at both ends caused by the magnetic end effect, therefore, needs to be considered from the design stage so as to minimize the unnecessary motor vibration and to increase the servo accuracy.

There exist various numerical and analytical methods in prior art to analyze the magnetic fields of LPMSMs including the finite element method (FEM), magnetic equivalent circuit (MEC) method, and subdomain analytical method (SAM). The numerical approach of FEM can be used to analyze motors with complex shapes and is considered the most accurate method [6], [7], [8]. However, a sufficient number of finite elements are required to achieve high accuracy [9], and thus, it is difficult to be used for motor design optimization due to the long calculation time. The MEC method divides analytical regions into several nodes and constructs a lumped magnetic circuit consisting of these nodes. It is relatively simple and has the advantage of being able to include magnetic field saturation [10], [11], [12], [13]. While the MEC modeling methods can notably reduce the calculation time, such methods obtain magnetic fields at only sparse points, thereby having relatively low model accuracy. The subdomain analytical method (SAM) directly find magnetic field solutions by solving the magnetostatic Maxwell equations and so achieve relatively high modeling fidelity with high speed [14], [15], [16]. Such methods, however, assume in general that a fundamental unit of LPMSMs repeats infinitely in the moving direction, which neglect the magnetic end effect.

Various attempts have been made to modify the SAM to include the magnetic end effect [17], [18], [19], [20], [21], [22]. In [18], [19], and [20], both ends of mover and stator are modeled by replacing the Cartesian coordinate with the cylindrical coordinate or adding a virtual unit at the longitudinal ends. The detent force obtained by these models shows high accuracy. Such approaches however do not model current regions, and therefore, additional force ripples driven by coil currents are not included. The current-driven force ripples, unlike the geometric-saliency-driven ripples (GSDRs), becomes significant as the coil current excitation increases, taking up a major portion of overall motor

force ripples. It is therefore important for a motor modeling method to include not only the geometric-saliency-driven force ripples but also the current-driven ripples (CDRs) in order to fully model the overall forces in LPMSMs. In [22], the suggested model contains a current region. However, they consider only the end of mover, limiting the application of such method to LPMSMs with multiple modular stators, which are widely used for the advantages of long-stroke, low cost, and ease of maintenance and repair [23]. The major contribution of this article is to propose a segmented layer modeling method to accurately capture both the geometric-saliency-driven force ripples and the current-driven force ripples of the LPMSMs with multiple modular stators. The proposed modeling method also simplifies the boundary conditions and improves the calculation efficiency by calculating solely a single modular force and superposing it to obtain overall motor force.

The rest of this article is organized as follows. In Section II, both the geometric-saliency-driven and current-driven force ripples caused by the magnetic end effect observed in an LPMSM with multiple modular stators are discussed. Section III presents segmented layer model (SLM) and its solution process. In Section IV, the segmented layer modeling method is validated as compared to an FEM model. The performance of SLM is experimentally validated using the testbed setup discussed in Section V. The associated experimental results are presented in Section VI. Finally, Section VII concludes this article.

II. FORCE RIPPLES IN LPMSM WITH MODULAR STATORS

A. Geometric-Saliency-Driven and Current-Driven Force Ripples

In this article, the force ripples are categorized into two major components, GSDR and CDR. The GSDR indicates the force ripple occurred by the motor geometric saliency with no current excitation. In the case of air-cored LPMSMs, the GSDR stems from the distortion of the magnetic field at the end of the movers and stators. Since such force ripple is not affected by the current excitation, it is a function of solely the relative position between the mover and the stator. Applying current to armature coils generates motor forces by the interaction between permanent magnets (PMs) and electromagnets, resulting in both the thrust and current-driven force ripples. The dc component and the ac component of current-generated motor force is defined as the motor thrust and the CDRs, respectively. Although both the GSDR and the CDR are affected by the magnetic end effects, only the CDR increases with the current excitation level, making the CDR as a function of both the current and the relative position of the mover to the stator.

B. Superposition of Motor Force

Fig. 1 shows a schematic of an LPMSM with a single PM mover moving over multiple modular stator armatures. In such an LPMSM configuration with multiple stators placed every period of Δ , the total motor force, including the GSDR and CDR, exerted on the PM mover can be obtained by the superposition of the modular force calculated within one period of Δ .

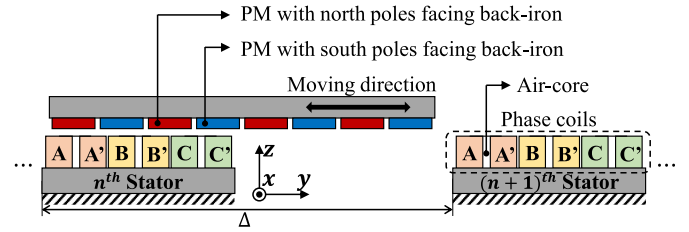


Fig. 1. Schematic of air-cored three-phase-four-magnet-combination LPMSM with multiple modular stator armatures periodically placed every Δ . Note that A , B , and C illustrate the phase coils, and their primed characters for the coil return paths.

Note that both the mover and modular stator have a finite length, and the longitudinal ends affect both the GSDR and CDR. The overall motor force is dependent on the relative position of the mover to the n th stator, denoted as y_r , and can be written as

$$F(y_r) = \cdots + F_{n-1}(y_r) + F_n(y_r) + F_{n+1}(y_r) + \cdots \quad (1)$$

where $F_n(y_r)$ represents the modular force generated by the n th stator armature module. When the stator modules are periodically placed by the distance of Δ , as shown in Fig. 1, $F_{n\pm 1}(y_r)$ is simply a phase-shifted version of $F_n(y_r)$ by $\pm\Delta$. This indicates that the motor force from any number of stators can be obtained by modeling an LPMSM with a single stator, such as the configuration shown within a single period of Δ in Fig. 1. Such an approach significantly simplifies model boundary conditions and reduces the model computation load and time. The following section therefore presents the analytical model of the LPMSM consisting of a single PM mover engaged with a single-stator module, which is capable of estimating the magnetic end effect of both the mover and the stator, and also of obtaining the overall motor force ripples, including both the GSDR and CDR, from an arbitrary number of modular stators.

III. SEGMENTED LAYER MODEL (SLM)

In this section, segmented layer modeling method is presented, which takes account of a current region and ends of both the mover and the stator. The proposed model divides the analytical region of the LPMSM into several segmented layers, hence called the SLM, and the magnetic field solutions are obtained by solving the Maxwell equations for the boundary conditions between layers.

A. Analytical Model of LPMSM

Analytical models of an air-cored LPMSM with a single-stator module in general have nine analytical regions to capture the ends of both the mover and stator, as shown in Fig. 2. For these regions with infinite lengths, applying the variable separation method to solve the Maxwell equations is restricted. Thus, the assumption that an LPMSM model is repeated with an arbitrary analytical length for each layer to have a finite spatial period is made so that magnetic field solutions can be obtained through the separation of variables. When the mover and the stator are repeated with a finite analytical length, the divided analytical regions 2-1 and 2-2 corresponding to the left and right

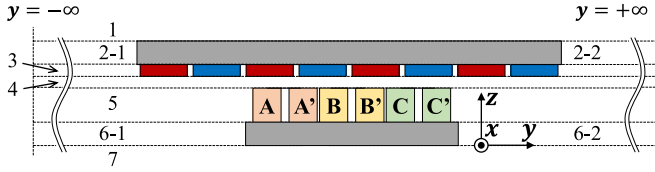


Fig. 2. Schematic of air-cored LPMSM with a single-stator module. Analytical regions are divided into nine layers, with infinite analytical length, considering both ends of the mover and the stator.

sides of the PM back-iron shown in Fig. 2 can be expressed as a single segmented layer (layer#2) to represent one side of the PM back-iron, as illustrated in Fig. 3. Likewise, the regions 6-1 and 6-2 in Fig. 2 can also be represented as a single segmented layer (layer#6) for one side of the coil back-iron, as shown in Fig. 3. Note that this approach does not reduce the accuracy of the solutions since the repeated motor configuration does not affect each other for a sufficiently long analytical length.

In SLM, therefore, the Maxwell equations are solved for a total of seven segmented layers, as shown in Fig. 3. The LPMSM model consists of a moving magnet track and a single-stator module, including a current layer, with the dimensions and coordinates as depicted in Fig. 3. In order to obtain magnetic field solutions for SLM, a few assumptions are made as follows.

- 1) Magnetic permeability of the soft magnetic material used for the PM and armature back-iron plates is infinite.
- 2) Magnetic permeability used in the PM layer (layer#3) is the same as that of a PM.
- 3) The third-dimension end effect of the LPMSM caused by the ends in the x -direction is not considered.

Poisson's equations [24] are solved for the magnetic fields in each segmented layer, written as

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J} - \mu_0 (\nabla \times \mathbf{M}) \quad (2)$$

where $\mathbf{A}$, μ_0 , $\mathbf{J}$, and $\mathbf{M}$ indicate the magnetic vector potential, permeability of air, current density, and magnetization vector, respectively. In a two-dimensional analytical problem, the vector potential $\mathbf{A}$ can be expressed using A_x alone, since it is the sole component perpendicular to the analytical plane. The Poisson's equations are obtained for each layer as

$$\nabla^2 A_{x,i} = \begin{cases} -\mu_0 (\nabla \times \mathbf{M}) & (i = 3) \\ -\mu_0 \mathbf{J} & (i = 5) \\ 0 & (\text{otherwise}) \end{cases} \quad (3)$$

where the index i indicates the layer number.

B. Particular Solution in PM Layer

For analytical model shown in Fig. 3, the magnetization vector $\mathbf{M}$ has only the z -direction component M_z and is a function of y alone; thus, it can be expressed in Fourier series as

$$M_z(y) = \sum_k (a_{mk} \cos pky + b_{mk} \sin pky) \quad (4)$$

where a_{mk} and b_{mk} represent Fourier series coefficients, and p denotes the fundamental spatial frequency, which can be written as $p = 2\pi/L_a$. Integrating both sides of $\nabla^2 A_{x,3} = -\mu_0 (\nabla \times$

$\mathbf{M})$ in terms of y , the z -component of the magnetic flux density corresponding to the particular solution in the PM layer can be obtained as

$$B_{z,3} = \mu_0 M_z(y) = \mu_0 \sum_k (a_{mk} \cos pky + b_{mk} \sin pky). \quad (5)$$

C. Particular Solution in Current Layer

In two-dimensional analytical model on the yz plane, as shown in Fig. 3, the current density $\mathbf{J}$ has only the x -direction component that points in and out of the plane, and it is a function of only y . The current density, which is the magnetic field source of the current layer, can be expressed in the form of Fourier series as

$$J_x(y) = \sum_k (a_{ck} \cos pky + b_{ck} \sin pky) \quad (6)$$

where a_{ck} and b_{ck} are the associated Fourier series coefficients. The z -component of the magnetic flux density corresponding to the particular solution in the current layer is then obtained by integrating both sides of $\nabla^2 A_{x,5} = -\mu_0 \mathbf{J}$ with respect to y , which can be written as

$$\begin{aligned} B_{z,5} &= \mu_0 \int J_x(y) dy \\ &= \mu_0 \sum_k \left(\frac{-b_{ck} \cos pky + a_{ck} \sin pky}{pk} \right). \end{aligned} \quad (7)$$

D. Homogeneous Solutions and Boundary Conditions

For all the layers in proposed model, the Laplace equation $\nabla^2 A_{x,i} = 0$ needs to be solved to obtain homogeneous solutions. By applying the separation of variables, each component of the magnetic flux density can be obtained as

$$\begin{aligned} B_{y,i} &= \frac{\partial A_{x,i}}{\partial z} = \sum_k \{ (\mathbb{L}_i(k) e^{pkz} - \mathbb{M}_i(k) e^{-pkz}) \cos pky \\ &\quad + (\mathbb{N}_i(k) e^{pkz} - \mathbb{O}_i(k) e^{-pkz}) \sin pky \} \end{aligned} \quad (8)$$

$$\begin{aligned} B_{z,i} &= -\frac{\partial A_{x,i}}{\partial y} = \sum_k \{ (\mathbb{L}_i(k) e^{pkz} + \mathbb{M}_i(k) e^{-pkz}) \sin pky \\ &\quad - (\mathbb{N}_i(k) e^{pkz} + \mathbb{O}_i(k) e^{-pkz}) \cos pky \}. \end{aligned} \quad (9)$$

Note that the solutions are expressed with the coefficients $\mathbb{L}_i$, $\mathbb{M}_i$, $\mathbb{N}_i$, and $\mathbb{O}_i$ in the form of Fourier series with its fundamental spatial frequency as p . However, layer#2 and layer#6 have shorter analytical lengths than other layers; thus, different fundamental spatial frequencies need to be applied. This is because the magnetic field inside the back-iron plate cannot be obtained following the assumption that the magnetic permeability of the soft magnetic material is infinite. The analytical lengths of layer#2 and layer#6, therefore, become $(L_a - L_r)$ and $(L_a - L_s)$, reduced by the length of each back-iron plate from the nominal analytical length L_a . The associated fundamental spatial frequencies of layer#2 and layer#6 are defined as $q = \pi/(L_a - L_r)$ and $r = \pi/(L_a - L_s)$, respectively, applying the Neumann boundary condition [17]. The variables p , k , and y ,

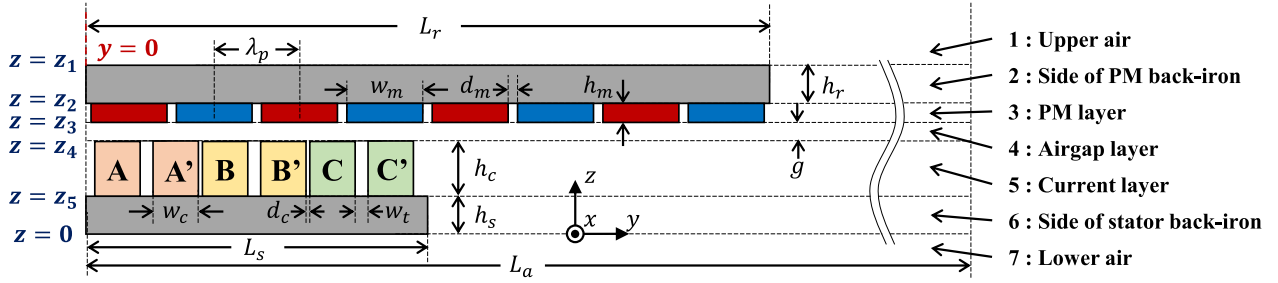


Fig. 3. SLM schematic of air-cored LPMSM with a single-stator module with reduced number of segmented layers. The model assumes that the mover and stator are repeated with analytical length of L_a . The mover, moving magnet track, consists of a back-iron plate with a total length of L_r and a thickness of h_r and alternately magnetized PMs with a width of w_m and a height of h_m , distanced to each other by d_m to have a pole-pitch of λ_p . Apart from the mover by the airgap of g , the model has a segmented layer of air-cored (core thickness of w_t) phase coils with a width of w_c and a height of h_c , distanced by d_c . The current layer is followed by the stator back-iron plate with a length of L_s and a thickness of h_s . Note that the y and z coordinates at the left-bottom end of the stator back-iron are defined to be zero. The vertical boundary locations for the layers are defined as z_1, z_2, z_3, z_4 , and z_5 .

TABLE I
ADJUSTMENTS IN MAGNETIC FIELD SOLUTION VARIABLES

Description	Layer#2	Layer#6
Fundamental spatial frequency (p)	q	r
Fourier series expansion number (k)	l	m
Coordinate (y)	$y - L_r$	$y - L_s$

used in (8) and (9), should be therefore replaced by the different variables for these layers, as organized in Table I.

SLM has a total of seven layers and the magnetic fields are expressed with a total of 28 coefficients from $\mathbb{L}_1, \mathbb{M}_1, \mathbb{N}_1, \mathbb{O}_1$ to $\mathbb{L}_7, \mathbb{M}_7, \mathbb{N}_7, \mathbb{O}_7$. It is however possible to reduce the number of coefficients by applying specific boundary conditions as follows.

- 1) The magnetic vector potential and magnetic flux density would go to zero as z approaches toward the positive or negative infinity. Thus, in layer#1, $\mathbb{L}_1$ and $\mathbb{N}_1$, which represent the coefficients of the exponentially growing function e^{pkz} , become zero. In layer#7 also, $\mathbb{M}_7$ and $\mathbb{O}_7$, the coefficients of the exponentially decaying function e^{-pkz} , should be zero for the same reason.
- 2) Considering the assumption that the magnetic permeability of the soft magnetic material is infinite, the magnetic field has only the normal components on the surfaces of the back-iron plates. Therefore, in layer#2 and layer#6, B_z equals zero at both y -direction ends of the back-iron plates, making the cosine term in (9) zero for these layers. Then, $\mathbb{N}_2$ and $\mathbb{O}_2$ become zero in layer#2, and $\mathbb{N}_6$ and $\mathbb{O}_6$ become zero in layer#6.

As a result, the magnetic field solutions of SLM are represented with a total of 20 coefficients, which can be obtained applying the boundary conditions for each layer as organized in Table II.

IV. MODEL FIDELITY VALIDATION WITH FEM

The fidelity of SLM is validated by comparing with the FEM. For an FEM model, the ANSYS Maxwell 2D software is used. The parameter values used for the model fidelity validation and experiments, discussed in Section VI, are organized in Table III.

TABLE II
BOUNDARY CONDITIONS FOR SLM

z -position	Boundary conditions
$z = z_1$	$H_{y,1} _{z=z_1} = \begin{cases} 0 & (0 \leq y \leq L_r) \\ H_{y,2} _{z=z_1} & (L_r \leq y \leq L_a) \end{cases}$ $B_{z,1} _{z=z_1} = B_{z,2} _{z=z_1} \quad (L_r \leq y \leq L_a)$
$z = z_2$	$H_{y,3} _{z=z_2} = \begin{cases} 0 & (0 \leq y \leq L_r) \\ H_{y,2} _{z=z_2} & (L_r \leq y \leq L_a) \end{cases}$ $B_{z,3} _{z=z_2} = B_{z,2} _{z=z_2} \quad (L_r \leq y \leq L_a)$
$z = z_3$	$H_{y,3} _{z=z_3} = H_{y,4} _{z=z_3}$ $B_{z,3} _{z=z_3} = B_{z,4} _{z=z_3}$
$z = z_4$	$H_{y,4} _{z=z_4} = H_{y,5} _{z=z_4}$ $B_{z,4} _{z=z_4} = B_{z,5} _{z=z_4}$
$z = z_5$	$H_{y,5} _{z=z_5} = \begin{cases} 0 & (0 \leq y \leq L_s) \\ H_{y,6} _{z=z_5} & (L_s \leq y \leq L_a) \end{cases}$ $B_{z,5} _{z=z_5} = B_{z,6} _{z=z_5} \quad (L_s \leq y \leq L_a)$
$z = 0$	$H_{y,7} _{z=0} = \begin{cases} 0 & (0 \leq y \leq L_s) \\ H_{y,6} _{z=0} & (L_s \leq y \leq L_a) \end{cases}$ $B_{z,6} _{z=0} = B_{z,7} _{z=0} \quad (L_s \leq y \leq L_a)$

TABLE III
PARAMETER VALUES USED IN MODEL FIDELITY VALIDATION AND EXPERIMENTS

Description	Symbol	Value
Width of mover back-iron	L_r	182.5 mm
Height of mover back-iron	h_r	10 mm
Width of PM	w_m	20 mm
Height of PM	h_m	5 mm
Relative permeability of PM	μ_r	1.099
Pole pitch	λ_p	22.5 mm
Airgap	g	1 mm
Width of coil	w_c	12 mm
Width of air-core	w_t	5 mm
Height of coil	h_c	15 mm
Coil pitch	λ_c	30 mm
Number of turns	N	314
Width of stator back-iron	L_s	92 mm
Height of stator back-iron	h_s	10 mm
Depth of motor (into the page)	D	60 mm
Analytical length (SLM)	L_a	900 mm

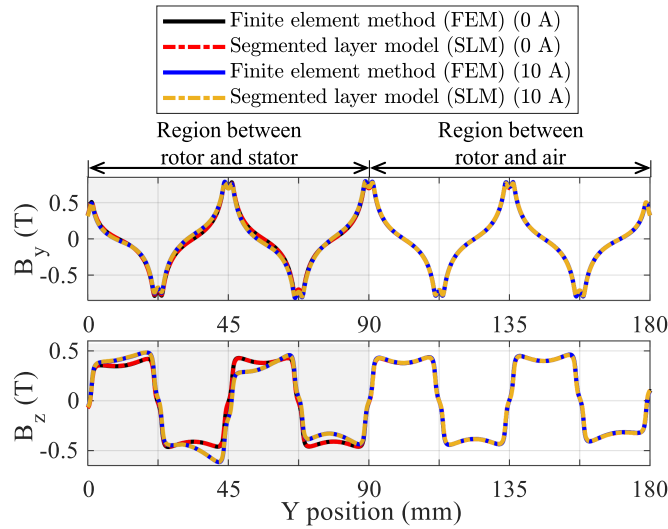


Fig. 4. SLM validation on B_y and B_z in the airgap, compared to FEM results. Magnetic fields are calculated both by SLM and FEM at a motor configuration shown in Fig. 3 when the left ends of mover and stator are aligned.

A. Magnetic Fields in Airgap

Magnetic field analysis results, comparing the fidelity of SLM to the FEM model, are shown in Fig. 4. Fields are calculated for both models at a motor configuration where the left ends of the mover and the stator are aligned, as shown in Fig. 3. The airgap fields B_y and B_z calculated at $z = (z_3 + z_4)/2$ from $y = 0$ to $y = L_r$ are compared in Fig. 4. For the excitation current of 0 A, the airgap magnetic fields are mostly periodic with nonperiodic magnetic end effects observed at both the left and right ends of the mover. When a phase current of 10 A is applied, the current-driven effect alters the airgap fields in the region where the PM mover is engaged with the armature stator module, as shown in the grey region in Fig. 4. The SLM captures both the magnetic end effect and the current-driven effect, showing significant agreement to the FEM results with NRMSEs (peak-to-peak-normalized root-mean-square errors) of 0.342% and 0.366% for B_y at 0 and 10 A, respectively, and 0.602% and 0.595% for B_z at 0 and 10 A, respectively. This validates the accuracy performance of proposed model, achieving more than 99% in estimating the airgap fields. Note that the SLM and the equivalent FEM model of the air-cored LPMSM stay in a linear operation range without a severe material saturation since the models have a relatively thick back-iron plate and a large airgap.

B. Motor Force by Multiple Modular Stators

Using the magnetic field solutions obtained by the SLM, the motor force can be calculated from the Maxwell stress tensor [24]. Motor force can be calculated for a range of mover positions to complete a motor force waveform against the relative position of the mover to the stator, where its zero position is defined as the configuration of Fig. 3 when the left ends of both mover and stator are aligned. The overall motor force from multiple modular stators can then be obtained by superposing

the modular motor force from each stator module distanced by a multiple of Δ . In this article, the LPMSM with one PM mover and two stator modules placed at the distance of $\Delta = 180$ mm is used, as shown in Fig. 5. Using the zero relative mover position defined above, *Position a* corresponds to $y_{a,1} = -90$ mm and $y_{a,2} = -270$ mm from the stator#1 and stator#2, respectively, while *Position b* is $y_{b,1} = 180$ mm and $y_{b,2} = 0$ mm, as illustrated in the figure. The modular motor force on the PM mover by each stator is calculated over a range of -270 to 180 mm.

The overall motor force ripples calculated by the proposed SLM are compared to the existing models in prior art, indicated as Model A [20], [21], Model B [14], [15], and Model C [22], and also to the FEM and the experimentally measured forces so as to validate the modeling fidelity performance. Fig. 6 shows the comparison results at the excitation current of 0 and 7 A as the PM mover moves from *Position a* to *Position b*, which is -90 to 180 mm referred from the stator#1. When the mover displacement is -90 to 0 mm and 90 to 180 mm, the PM mover is engaged solely with one stator module (stator#1 and stator#2, respectively), which is defined as an one-to-one engagement region. When the mover displacement is 0 to 90 mm, on the other hand, the mover is engaged both with the stator#1 and the stator#2, and it is defined as an one-to-two engagement region. In the one-to-one engagement regions, the GSDRs are dominant, showing relatively small changes when applying a high current excitation of 7 A. Note that Models B and C assume an infinitely long armature stator, limiting in calculating the force ripple by the magnetic end effect at the no-load condition. The force ripple estimation results of the SLM are therefore compared to Model A and the FEM at 0 A, as shown in Fig. 6(a). While Model A, assuming an infinitely long PM track, captures only the mover-oriented GSDR, as shown in the figure, the proposed SLM accurately captures the ripple components generated by the ends of the modular stators, significantly agreeing with the FEM results.

In the one-to-two engagement region, on the other hand, the current-driven force ripples are dominant, showing significant ripple increase when a current of 7 A is applied. This implies that in an LPMSM with multiple modular stators, the motor force is significantly influenced by the current-driven end-effect ripple as the PM mover moves across the multiple stator modules at the same time. Such current-driven end-effect ripples are not captured by Model B, as shown in Fig. 6(b), since it assumes the periodicity for both the mover and the stator. It is also observed in the figure that Model C significantly underestimates the force ripple due to the assumption of the infinitely long stator. On the other hand, the SLM captures both ends of the mover and the stator modules even on the on-load condition, showing significant agreements with the FEM results both in the one-to-one and one-to-two engagement regions. Fig. 6 also shows that the proposed SLM significantly agrees with the experimental results as compared to the other existing models. The experimental validation results on the force ripple estimation performance of the proposed SLM are discussed in depth in Section VI-B. Note that the motor force of both the models and the prototype of the air-cored LPMSM shows a linear behavior for a wide range of phase current excitation, including the current level of

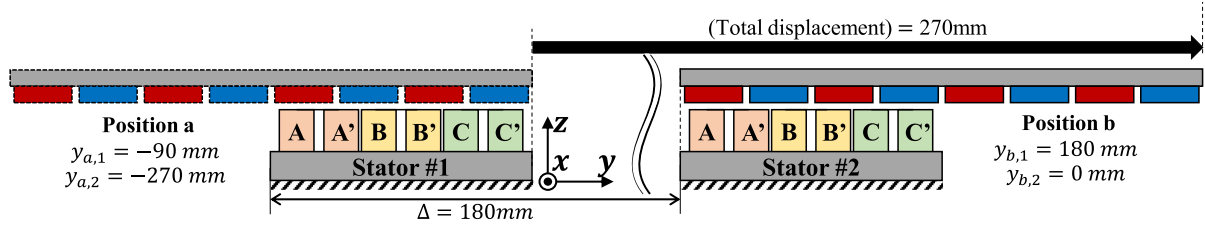


Fig. 5. Schematic of LPMSM configuration with two short-armature stator modules, distanced by $\Delta = 180$ mm, and one PM mover moving from *Position a* to *Position b*. From the stator#1, the relative mover positions at *Position a* and *Position b* correspond to $y_{a,1} = -90$ mm and $y_{b,1} = 180$ mm, respectively. From the stator#2, the relative mover positions are likewise $y_{a,2} = -270$ mm and $y_{b,2} = 0$ mm. Modular motor force on the PM mover from each stator module is calculated for the range of -270 to 180 mm.

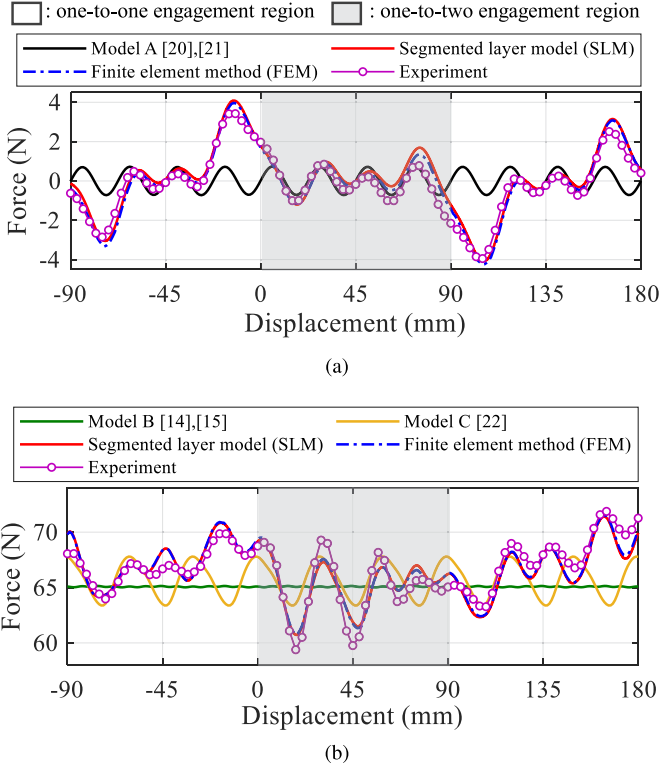


Fig. 6. Model fidelity validation of the proposed SLM as compared to the prior models [14], [15], [20], [21], [22], FEM, and experimental results, in terms of motor force ripples at (a) 0 A and (b) 7 A. The SLM shows significant agreements with both FEM and experimental results, as compared to the previous methods, both at a zero-current case and at a high current excitation of 7 A. Note that the testbed setup and the experimental validation results are discussed in depth in Sections V and VI, respectively.

TABLE IV
COMPARISON IN SIMULATION RESULTS OF EACH MODEL TO FEM

	0A			7A			
	Model A	SLM	FEM	Model B	Model C	SLM	FEM
NRMSE (%)	20.265	2.200	—	25.470	24.649	1.317	—
Thrust (N)	—	—	—	65.079	65.715	66.427	66.473

The bold values to emphasize the modeling accuracy of the proposed SLM.

7 A, due to the thick back-iron plate and the relatively large airgap.

The comparison results are organized in Table IV in terms of the NRMSE and the thrust, quantitatively showing the significant model fidelity of the proposed SLM, as compared to the other

TABLE V
CALCULATION TIME COMPARISON IN MOTOR FORCE ANALYSIS

Model A	Model B	Model C	SLM	FEM
0.6 s	0.05 s	0.6 s	0.8 s	80 s

The bold values to emphasize the modeling accuracy of the proposed SLM.

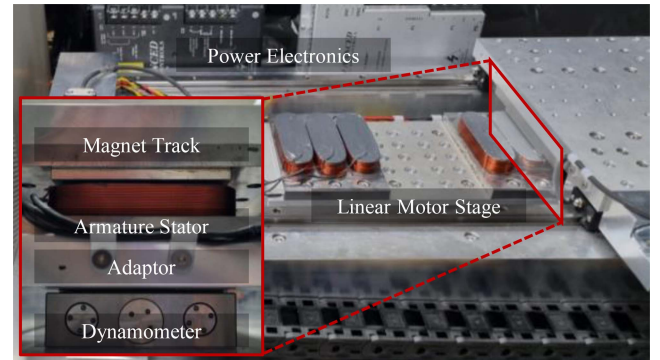


Fig. 7. Experimental testbed to validate the fidelity of the proposed SLM. The three-phase LPMSM, consisting of a moving magnet track and two air-cored modular stators, is used to drive the linear stage. The motor forces are measured by the dynamometer installed below the stator module through the adaptor, as shown in the magnified view.

prior art. The calculation time required for the motor force analysis by each method, for instance at a single point in Fig. 6, is summarized in Table V. The calculation time of the SLM stays close to the other prior models while showing two orders of magnitude lower as compared to the FEM using the identical computing system with a 6-cored 3.60 GHz CPU and a 32 GB RAM.

V. EXPERIMENTAL TESTBED

Fig. 7 shows the linear motor stage testbed constructed to experimentally validate the fidelity of the SLM. The stage is driven by the LPMSM, with two air-cored three-phase armatures as the short stator modules and the magnet track installed underneath the moving stage as the single mover. The two modular stators are fixed onto the optical table through an adaptor where the dynamometer (Type 9129AA by Kistler) is installed in-between, as shown in the magnified view in the figure. The motor phase coils are driven by the pulsewidth modulation amplifiers (B30A40 by Advanced Motion Controls), which are powered by the power supply unit (PS30A by AMC). An optical encoder (Ti0020A08A for the interface, RTCL-S for the scale,

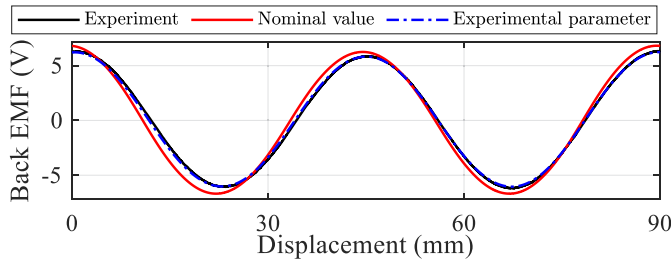


Fig. 8. Back-EMF curves of phase B coils obtained from experiment and SLM using the nominal design values and the identified experimental parameters in the one-to-two engagement region.

and T1031 for the read-head by Renishaw) with a resolution of $1\ \mu\text{m}$ is used to determine the stage position. The motor forces are directly measured by the dynamometer installed under the stator modules, and the output voltages are taken into the FPGA module (PXIe-7856R by National Instruments) to monitor the real-time force measurements, which are used to validate the motor-force-estimating performance of SLM.

VI. EXPERIMENTAL VALIDATION

A. Experimental Motor Parameters

When it comes to fabricating an LPMSM and a linear stage, the manufacturing and assembly errors can alter the motor performance. Considering such an issue, the experimental motor parameters of testbed are obtained by taking in the fabrication errors to perform more realistic and accurate validation of SLM. Fig. 8 shows the back-EMF curves of the phase B coils in stator modules obtained by the experiment and also from SLM in the one-to-two engagement region. Using the nominal design values in Table III in SLM, differences in the magnitude and the phase of the back-EMFs are observed, compared to the experimental result. Such discrepancy stems from the fabrication errors, altering the magnetic fields in the airgap and the associated magnetomotive force harmonic components, affecting the back-EMF and hence the motor forces.

In order to take such fabrication tolerances into account, the design parameters such as the distance between PMs, the magnitude of PM magnetization, and the number of turns of the phase coils are modified, considering the experimental hardware. Using such experimental motor parameters, SLM calculates the back-EMF significantly closely to the experimental data, as shown with the blue dash-dotted line in Fig. 8. These experimental parameters are used to calculate the force ripple of the LPMSM testbed using the proposed model, and compare the results with the experimental force measurement.

B. Force Ripple Validation by Experiment

Fig. 9 shows the force ripple estimation performance of SLM as compared to the FEM and experimental results, using the experimental motor parameters discussed in Section VI-A. At the current excitation of 0 A, as in Fig. 9(a), SLM predicts the cogging force significantly closely to the FEM and the experimental measurement with the NRMSE of 0.885% and 5.022% in the one-to-one engagement region, and 3.696% and

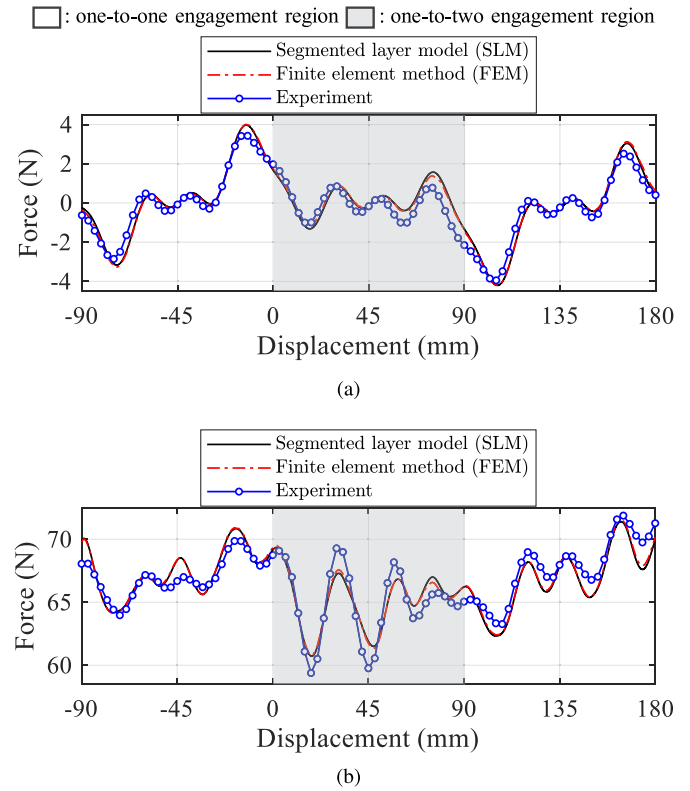


Fig. 9. Force ripple estimation performance of SLM compared to FEM and experimental results at the phase current of (a) 0 A and (b) 7 A, showing significant agreements. The dc thrust values at 7 A are organized in Table VI.

TABLE VI
COMPARISON OF SIMULATION RESULTS (SLM AND FEM) TO
EXPERIMENTAL MEASUREMENTS

		0A		
		SLM	FEM	Experiment
one-to-one engagement	peak-to-peak (N)	7.236	7.331	6.501
	NRMSE (%)	-	0.885	5.022
one-to-two engagement	peak-to-peak (N)	3.017	3.100	4.126
	NRMSE (%)	-	3.696	12.403
		7A		
		SLM	FEM	Experiment
one-to-one engagement	peak-to-peak (N)	9.076	9.149	8.800
	NRMSE (%)	-	0.984	10.233
one-to-two engagement	peak-to-peak (N)	8.627	8.784	10.133
	NRMSE (%)	-	2.215	11.776
Thrust (N)		66.427	66.473	66.558

12.403% in the one-to-two engagement region, respectively. These comparison results are organized in Table VI.

As a higher current excitation of 7 A is applied, the force ripple as well as the thrust is generated, especially severe in the one-to-two engagement region, as shown in Fig. 9(b). Compared to the force ripple at 0 A, which is the cogging, the ripple peak-to-peak at 7 A in the one-to-two engagement region is increased by 146% from 4.1 to 10.1 N while it is 35% in the one-to-one engagement region from 6.5 to 8.8 N. This indicates that the current-driven end-effect force ripple becomes more significant when the PM mover moves over the multiple stator modules

at the same time as discussed in Section IV-B. Even for such situations with significant current-driven end effects existing, the motor force estimation by SLM agrees with the FEM and the experiment with the NRMSE of 0.984% and 10.233% in the one-to-one engagement, and 2.215% and 11.776% in the one-to-two engagement, respectively, as shown in Table VI. The thrust estimation of SLM also significantly agrees both with the FEM and the experimental measurement at around 66 N at 7 A, with the error of less than 1% as can be seen from the table. Note that considering the advantages of such high-fidelity force estimation and rapid calculation speed, the proposed modeling method can be readily utilized in the design optimization process of, for instance, low-ripple air-cored LPMSMs for high-precision servo control systems.

The discrepancy observed in the force ripples at 7 A between the simulation (SLM and FEM) and the experiment might come from the fabrication errors such as the height differences of the PMs and phase-coil windings. Such parameter differences are not taken into account for the proposed SLM since it would require additional layers, thereby complicating the analytical model and increasing the computational time. There is also the end effects in the motor-depth-direction contributing to the discrepancy since the SLM and the FEM are in the two-dimensional space. In addition, the three-dimensional effects from the coil end-turns can also cause the difference on the on-load condition, which can be observed that the discrepancy between the two-dimensional simulation and the experimental data becomes larger when the nonzero phase currents are applied, especially in the one-to-two engaging region, as shown in Fig. 9.

VII. CONCLUSION

This article presented the SLM, which divides the motor structure into segmented layers to capture the force ripples caused by magnetic end effects of LPMSMs with multiple modular stators. Using the proposed SLM, the CDRs as well as the GSDRs can be accurately estimated for the LPMSMs with multiple modular stators. The SLM showed an accuracy of more than 97%, compared to the FEM, while reducing the calculation time by two orders of magnitude. The performance of the SLM was also experimentally validated. The SLM estimated the current-driven force ripples, especially significant when the PM mover moves across multiple stator modules at the same time, with an accuracy of more than 87% as compared to the experimental force measurements. The motor thrust obtained by the SLM showed significant agreement both with the FEM and the experiments within the error of 1%. The manufacturing and assembly errors can be a major cause of the discrepancy between the SLM and the experiments. When a precise motor force estimation is required, such as for the application of force ripple reduction controls, the overall force ripples in the LPMSM can be accurately estimated through an equivalent model in which the fabrication errors were taken into account. Such equivalent model, which is one of our future research works, can be established in an iterative manner, sweeping the tolerance ranges of the motor parameters. The proposed SLM can be

readily used for that purpose due to its high fidelity and short computation time.

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